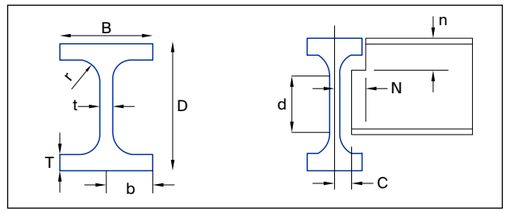
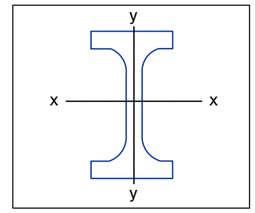
2.4
Metric Beams & Columns (MB & MC)



Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm



Metric Units

Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre			Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
400x400	140	388	402	15	15	22	314	178	13.4	20.9	10	204	37
	147	394	398	18	11	22	314	187	11.1	28.5	8	204	40
	172	400	400	21	13	22	314	219	9.52	24.2	9	204	43
	197	400	408	21	21	22	314	251	9.71	15.0	13	204	43
	200	406	403	24	16	22	314	255	8.40	19.6	10	204	46
	232	414	405	28	18	22	314	295	7.23	17.4	11	204	50
	235	408	412	25	25	22	314	300	8.24	12.6	15	204	47
	283	428	407	35	20	22	314	361	5.81	15.7	12	204	57
	415	458	417	50	30	22	314	529	4.17	10.5	17	204	72
	605	498	432	70	45	22	314	770	3.09	7.0	25	204	92
450x200	65.1	446	199	12	8	13	396	83.0	8.29	49.5	6	106	25
	66.2	446	199	12	8	18	386	84.3	8.29	48.3	6	106	30
	74.9	450	200	14	9	13	396	95.4	7.14	44.0	7	106	27
	76.0	450	200	14	9	18	386	96.8	7.14	42.9	7	106	32
	88.9	456	201	17	10	18	386	113	5.91	38.6	7	106	35
	98.9	460	202	19	11	18	386	126	5.32	35.1	8	106	37
	110	460	205	19	14	18	386	140	5.39	27.6	9	106	37
450x300	106	434	299	15	10	24	356	135	9.97	35.6	7	155	39
	121	440	300	18	11	13	378	154	8.33	34.4	8	155	31
	124	440	300	18	11	24	356	157	8.33	32.4	8	155	42
	145	446	302	21	13	24	356	184	7.19	27.4	9	155	45
500x200	77.9	496	199	14	9	13	442	99	7.11	49.1	7	105	27
	79.5	496	199	14	9	20	428	101	7.11	47.6	7	105	34
	88.2	500	200	16	10	13	442	112	6.25	44.2	7	105	29
	89.6	500	200	16	10	20	428	114	6.25	42.8	7	105	36
	102	506	201	19	11	13	442	129	5.29	40.2	8	105	32
	103	506	201	19	11	20	428	131	5.29	38.9	8	105	39
	117	512	202	22	12	20	428	148	4.59	35.7	8	105	42
500x300	111	482	300	15	11	13	426	141	10.0	38.7	8	155	28
	114	482	300	15	11	26	400	146	10.0	36.4	8	155	41
	125	488	300	18	11	13	426	159	8.33	38.7	8	155	31
	128	488	300	18	11	26	400	164	8.33	36.4	8	155	44
	150	494	302	21	13	26	400	191	7.19	30.8	9	155	47

Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
2.32	48970	16260	16.6	9.55	2524	809	2802	1237	0.830	22.5	5.66	156
2.32	56150	18930	17.3	10.1	2850	951	3118	1441	0.850	20.9	6.69	194
2.34	66620	22420	17.5	10.1	3331	1121	3672	1700	0.850	18.2	8.05	303
2.35	70890	23810	16.8	9.75	3545	1167	3992	1794	0.828	16.7	8.55	415
2.35	78040	26200	17.5	10.1	3844	1300	4280	1977	0.846	16.1	9.56	462
2.37	92770	31030	17.7	10.2	4482	1532	5026	2331	0.848	14.1	11.6	714
2.38	86470	29200	17.0	9.87	4239	1418	4818	2185	0.827	14.2	10.7	694
2.41	119200	39360	18.2	10.4	5570	1934	6311	2941	0.854	11.6	15.2	1317
2.49	187100	60530	18.8	10.7	8172	2903	9540	4436	0.853	8.5	25.2	3885
2.60	297900	94360	19.7	11.1	11960	4369	14460	6724	0.852	6.4	43.2	11060
1.65	28130	1579	18.4	4.36	1262	159	1423	245	0.876	38.4	0.744	33.9
1.64	28700	1581	18.5	4.33	1287	159	1450	247	0.878	36.5	0.744	38.3
1.66	32890	1870	18.6	4.43	1462	187	1652	290	0.878	33.6	0.889	51.6
1.65	33450	1872	18.6	4.40	1487	187	1679	291	0.879	32.2	0.890	56.9
1.67	40400	2308	18.9	4.51	1772	230	2003	356	0.884	27.5	1.11	92.5
1.68	45430	2619	19.0	4.56	1975	259	2240	403	0.885	25.0	1.27	126
1.68	47860	2742	18.5	4.43	2081	268	2399	423	0.869	23.9	1.33	152
2.00	46800	6695	18.6	7.04	2157	448	2384	686	0.883	27.0	2.94	104
2.04	54730	8106	18.9	7.26	2488	540	2757	823	0.885	24.9	3.61	141
2.02	56070	8114	18.9	7.18	2549	541	2825	828	0.887	23.4	3.61	163
2.03	66380	9658	19.0	7.24	2977	640	3323	981	0.886	20.5	4.36	253
1.75	40840	1843	20.3	4.31	1647	185	1869	288	0.874	37.5	1.07	52.5
1.74	41870	1846	20.3	4.27	1688	186	1914	290	0.876	35.2	1.07	60.8
1.76	46810	2138	20.4	4.36	1872	214	2130	333	0.875	33.3	1.25	75.9
1.75	47850	2142	20.5	4.33	1914	214	2175	335	0.877	31.6	1.25	85.9
1.77	55480	2578	20.7	4.47	2193	257	2496	399	0.880	28.7	1.53	119
1.76	56520	2582	20.7	4.43	2234	257	2541	401	0.881	27.5	1.53	132
1.77	65450	3035	21.0	4.52	2557	300	2913	469	0.884	24.3	1.82	193
2.12	58270	6756	20.3	6.92	2418	450	2696	690	0.877	32.4	3.68	93.9
2.10	60370	6768	20.4	6.82	2505	451	2791	695	0.880	29.4	3.69	118
2.13	68860	8106	20.8	7.14	2822	540	3132	825	0.887	28.0	4.48	143
2.11	70960	8118	20.8	7.05	2908	541	3228	830	0.889	25.9	4.48	172
2.13	83810	9663	20.9	7.11	3393	640	3792	984	0.888	22.8	5.40	265